

Educational Computer Game for learning History: Impact on learners' level of knowledge and learners' satisfaction

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Abstract— Educational computer games have been adopted, for a long time, as a valuable tool in the learning and teaching of several subjects such as physics concepts, science, reading and spelling. These games are considered as a powerful instrument for teaching as they succeeded in improving learners' motivation, increase learners' desire to learn, make learning more enjoyable, and lead learners to complete more difficult work. However, the use of these games for learning also has potential to affect the learning experience negatively if the game designer does not balance between a set of critical factors. Generally, the right combination of fun gameplay and appropriate learning experience constitutes a good opportunity for creating an effective educational game that supports the learners to acquire knowledge and to increase their competency. In this context, this paper introduces and describes a new educational computer game, called "LAST MISSION", especially designed for learning the history of the World War 1 and the World War 2. In addition, this work seeks, through a pilot study, to investigate whether the use of this educational computer game has a positive effect on learners' level of knowledge and learners' satisfaction.

Keywords— Educational computer game, Learning tools, History; Level of knowledge; Technology Acceptance Model

I. INTRODUCTION

Using computer games in education is a new manner of exploiting technologies to enhance the learning activity. These games are developed for the purpose of making learners having fun and discovering new knowledge. Educational computer games can offer several learning benefits to learners. In fact, the use of educational games can consume the learner's attention and increase his/her engagement and motivation which could lead to stimulate learning. In addition, these games provide player with tactile, visual and intellectual stimulation; Expand and stimulate creativity; Allow real collaboration between players; develop self-confidence and assist selfhelp and social skills through collaborative working [1]. Furthermore, a growing number of research studies [2, 3, 4, 5, 6] show that learnt through playing has been proven to be successful learning experience. Moreover, many researchers

(e.g. [2, 3, 4]) argue that computer educational games can be more interesting, more enjoyable and thus more effective than traditional learning modes to improve the knowledge of the learners. Other researchers [5, 6] argue that educational computer games can solve the problems of traditional learning by enhancing learning, engaging learners and providing studying methods that correspond better with learners' requirements and habits. Therefore, it is desirable to use educational computer games for teaching and learning.

In this context, the paper introduces and describes a new educational computer game, called "LAST MISSION", especially designed for learning the history of the World War 1 and the World War 2. The game was properly implemented using high graphic 2D platform design and presents a military environment style as it is developed for learning History of World Wars. The game can be classified under the action genre but without any violence scenes. The game also provides instantaneous feedback for the learner, so he/she can use them to determine how well he/she understand what he/she is trying to learn.

This paper also presents an experimental study, conducted with volunteer participants. Precisely, a total of 36 participants from different schools in Kairouan (Tunisia) voluntarily participated in this study for evaluating the effectiveness of this educational computer game. In particular, this work seeks to investigate whether the use of this educational computer game has positive effects on learners' level of knowledge and learners' satisfaction.

The reminder of this paper is organized as follows: the next section starts by presenting a definition of the three main concepts and provides an overview of related to the use of the computer educational games. Section 3 introduces the "LAST MISSION" game and describes the main features and levels of this game. Section 4 describes an experimentation conducted in order to evaluate the effectiveness of this game. Finally, section 5 concludes the paper with a summary and future works.

II. RELATED WORKS

A. Game and play

The first question considered as part of this work is definition of the term game. Khenissi et al. [4] argued that it is difficult to come up with a just one definition of a game. For that, several selections of definitions are considered here.

Roger Caillois [7], in his book (Games and Men), described the game as an activity that should have the following criteria:

- fun: the activity is selected for its light-hearted character
- separate: it is restricted in place and time
- uncertain: the result of the activity is unexpected
- non-productive: player does not achieve anything useful
- controlled by rules: the activity has several rules different from everyday life
- fictitious: it is accompanied by the consciousness of different reality.

Klabbers [8] describes games as “an activity or sport involving skill, knowledge or chance, in which you follow fixed rules and try to win against an opponent to solve a puzzle.”

Oxland [9] says that games need boundaries and rules, an interface to the game world, context sensitivity (or immersion), goals, feedback, challenges and quests, a balance and game environment.

Koster [10] provides a less formal definition. Koster says that games are puzzles to be solving and exercises for the brains. He also says that it is the act of solving these puzzles that makes games fun.

Ehrmann [11] provides the following definition of play:

“From the standpoint of form, we can define play in short as a free activity, experienced as ‘make-believe’ and situated outside of everyday life, nevertheless capable of totally absorbing the player; an activity entirely lacking in material interest and inutility”.

B. Educational computer games

In recent years, game studies have rapidly developed into educational computer games. Van Eck [12] defined these games as an interactive play that teaches learners how to achieve goals and how some serious problems can be solved.

Van Eck [12] considered that educational computer games have fantasy elements that could engages learners in a learning activities through story-lines or narrative where learners are able to express themselves as individuals while engaging and learning in social issues. Kiili [13] have considered that educational computer games are magic tools that could solve the issues of computer-assisted learning by enhancing learning, engaging learners and providing studying methods that correspond better with learners’ requirements and habits. Van Eck [12] says that educational computer game is a game explicitly designed with learning purposes, where the role of fun is obvious and required.

Using this type of games in studying can lead to enjoyable and effective learning experience that will help learners to study about certain subjects, expanding concepts, reinforcing learning and development skills. Because of its engaging nature, educational computer games give the ability to learners to enter a state of relaxing and enjoying where they are very receptive to learning materials [14].

According to several researchers (e.g [15], [16], [17], [18] [19], [20], [21]), the usage of educational computer games in the educational process has a high level potential, which resides in the degree of motivation involved playing act: making advance in the investigation and assimilating of new knowledge. These researchers stressed that there are many advantages of using educational computer games. Some of these advantages are listed below:

- Educational computer games can provide elements of interactivity that may stimulate learning. According to Deubel [22], educational computer games have the potential to motivate and engage learners, and offer good learning experience.
- Educational computer games allow learner to experience novelty, curiosity and challenge. All these factors may stimulate learning.
- Educational computer games may help in the development of cognitive skills. Indeed, educational computer games are a new challenge to the teaching community. Educational games provide the development of cognitive skills in particular and making possible the socio-cultural interactions.
- Educational computer games can assist learners in setting goals, preserve records of behavioural change, ensure learning goals and provide feedback.
- Educational computer games may help learners in developing computer skills needed in the society that continues to expand technologically.
- Educational computer games help in increasing learner's engagement, performance, competency and skills while gaming; and lead to an enjoyable and effective learning experience.

III. "LAST MISSION": AN EDUCATIONAL COMPUTER GAME FOR LEARNING HISTORY

This section describes an educational computer game, called “Last Mission”, especially designed for learning History. The game was developed under GML programming language and exported for HTML 5 browsers.

The game was implemented using high graphic 2D platform design and presents a military environment style as it is developed for History studying. The game can be classified under the action genre but without any violence scenes.

The interface of the game is very accurate and well adapted for its objective, which can attracts players ‘eyes and provides an immersive gaming experience. The game perspective contains different details like timing and score at the top left corner and the current level at the bottom left

corner. The health bar and the bonus indicator are situated on the top of the interface. The game play is quite easy and simple and does not need much effort from players, but it still fun and enjoyable.

The protagonist of the game is a military soldier where his objective is to shoot the front enemies and to gain score and bonus items. In the LAST MISSION game, bonus items are generally information about important events from the history of World War 1 and World War 2.

This game consists of three main levels which are: (1) Training mission, (2) Facing enemy soldiers and (3) Facing enemy tanks and air planes.

A. Level 1: Training mission

There are three different targets in this level. Each target contains a number between 1 and 3. In the left bottom corner of the room interface, a multiple choice question is proposed, when the player answers the current question, the next question will be generated automatically. For each proposed question there are 3 propositions with numbers from 1 to 3. Only one proposition is correct. Each proposition has a moving target with the same number. When the player decides the correct proposition he/she must shoot the corresponding target with the same number of the chosen proposition. In case of a correct answer, +1 will be added to the total number of correct answers otherwise it will be added to the total number of wrong answers. In the right top corner of the room interface, there are two icons, the green one shows the number of correct answers and the orange one shows the number of wrong answers. After completing all proposed questions the game will load the next level. Figure 1 shows the interface of the level 1.

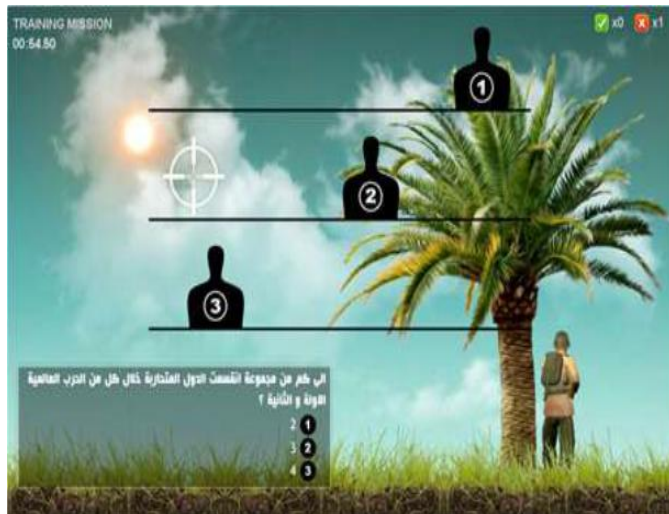


Fig.1 Interface of the level 1

B. Level 2: Facing enemy soldiers

In this level the player needs to control a soldier character that is able of shooting bullets using a rifle on his hands. The player's objectives behind this mission are to face enemy soldiers and to collect a lot of bonus items. When completing one of the objectives, the player will gain new information about events from the World War 1. An orange box with

exclamation point in game interface represents a bonus item. When the player collides with those boxes a dialogue popup will appears above the controlled character and shows some useful information. After finishing reading, the player can hide this dialogue and completes the mission. When the player reaches the end of this level, five questions will appear with another dialogue popup. Each question has three different propositions with yellow backgrounds. In case of selecting the wrong answer, the background of the chosen proposition will change into red colour, otherwise it will change into green. Figure 2 presents the interface of the level 2.



Fig.2 Interface of the level 2

C. Level 3: Facing enemy tanks and air planes

In this level, the player controls a military car able of shooting big bullets from a weapon situated above it. All faced air planes and tanks represent the enemy's army. The player can defeat them and destroy each faced air plane or tank using his weapon. The same case in this level as for the previous one, each collected bonus item generates important information, but this time it will be about some events from the history of World War 2. In addition, when the player reaches the end of this level he/she must answers five new questions with three propositions. Figure 3 presents the interface of the level 3.



Fig.3 Interface of the level 3

In this game, each level starts with a dialogue box that contains useful game instructions. The score points represent the total number of killed soldiers or destroyed air planes and tanks and the red bar in the top indicates the player's current health statue.

In summary, this game contains three different levels with distinct goals. The player must finish all of them in order to complete the game's missions. Playing this game is quite simple so that young players could easily start the game with any assistance.

IV. EXPERIMENT DESIGN

This section describes an experimentation conducted for evaluating the effectiveness of this educational computer game. In particular, this work studies the impact of this educational computer game on learners' satisfaction and learners' level of knowledge.

A. Participants

This experiment involved 36 participants from different schools in Kairouan (Tunisia) voluntarily participated in the study. Participants were affected randomly to two sub-groups. The first group includes 20 participants that will learn the course history of World War 1 and World War 2 in traditional way. The second group includes 16 participants. These participants will learn the same course (slightly modified) using the educational computer game. Table 1 shows the number of participants; number of Female and Male participants; and the average age of both first group and second group participated in this experimental study.

TABLE 1: PARTICIPANTS IN THE EXPERIMENTAL STUDY

	Total	Male	Female	Avg. Age
First group	20	15	5	20
Second group	16	13	3	18
Total	36	28	8	19

Table 1 shows that the percentage of learners that participated in the first group of the experimental study is about 56%, while for the second group, the percentage of learners that participated is about 44%. In addition, the percentage of female participants in both groups is about 22% (14% in the first group and 8% in the second group). Moreover, the average age recorded for the first group is 20 while the average age recorded for the second group is 18. From this table, we could conclude that the two groups are relatively balanced

V. PROCEDURE

As mentioned in the previous sub-section, this experiment involved two groups. For both groups, a pre-test and a post-test were used in order to evaluate the learners' achievements. The pre-test is used in order to evaluate learners' basic knowledge about important events of the history of World

War 1 and World War 2, while the post-test is used in order to discover how much learners does acquire new knowledge after the learning using traditional learning and educational computer game.

The pre-test and the post-test were well prepared by an instructor of history. Each test was composed of 10 different multiple choice questions about the studied subject. Concerning the evaluation, the learner will get 1 point if only he/she selected the correct answer, and no subtractions points for wrong answers. In summary, the total points that each learner possibly can get is restricted between a maximum of 10 points and a minimum of 0 points.

1) Phase 1: traditional learning

The proposed learning methodology in this phase is the traditional one. The history course of World War 1 & 2 was printed into papers (4 pages) and given to participants of the first group.

Some of the participated learners prefer reading the course using a computer and others prefer reading from printed papers. Before starting reading the course, participants were invited to answer a pre-test composed of 10 multiple choice questions. After that, the participants take about 20 minutes in reading the provided course. After finishing reading, participants of the first group were invited to answer the post-test questions. Finally, results of both tests were saved.

2) Phase 2: Learning by using educational computer game

This phase was reserved for evaluating the impact of sing the proposed game. This phase uses LAST MISSION computer game for teaching participants the course history of World War 1 and World War 2.

As in the previous phase of this experimental study, two tests were provided to the participants. The first one is a pre-test provided to participants before starting playing the game and the second one is a post-test after finishing the game. It is important to note that both tests include the same multiple choice questions. After having used the educational computer games, the averages of participants' scores in the pre-test and the post-test were calculated.

3) Learners' satisfaction when using the game

After completing the both tests, participants of the second groupe were invited to answer a Technology Acceptance Model (TAM) questionnaire [23]. Specifically, participants of the second group (experimental group) were invited to answer to a validated questionnaire. The questionnaire is developed to determine the learner satisfaction when using the computer educational game for learning. The questionnaire is composed of the 3 items for Usefulness (U) and the 3 items for Ease Of Use (EOU). The questionnaire contains also instances of the 4 items for attitude toward using the system (ATT) and the 3 items for behavioural intention to use the system (INT) presented in [23]. In addition, each item of the questionnaire contains 7 values on Likert scale ranging from 1 which represents the total agreement to 7 representing the total disagreement.

VI. RESULTS

1) Impact of the educational computer game on the learners' level of knowledge

For evaluating the impact of this educational computer game on the learners' level of knowledge, this study used a pre-test and a post-test and two sub groups. Table 2 shows all results obtained from phase 1 and phase 2 of the experimental study. The first phase presents the traditional method for learning the history course while the second phase presents the new way to learner which is by using the educational computer game. Table 2 shows that the average value obtained from the pre-test of the first phase is equal to 5.65 and for the post-test is equal to 7.85. This means that the progression rate obtained is equal to 2.2. While for the second phase, the average value from the pre-test is equal to 5.25 and for the post-test is equal to 8.37. In this case the obtained progression rate is positive and equal to 3.12. Table two shows the results of the Wilcoxon test [24].

TABLE 2: RESULTS OF THE WILCOXON TEST

	Test	Average	W-Value	p
Phase 1	Pre-test	5.65	9	$p \leq 0.01$
	Post-test	7.85		
Phase 2	Pre-test	5.25	0	$p \leq 0.01$
	Post-test	8.37		

Table 2 presents the Wilcoxon test which is a nonparametric test specifically designed for evaluating the difference between 2 conditions or treatments where the samples are correlated. This test is based on a signed ranking for calculating a W-value used for determining a significance degree (p). The p-value will be compared to a defined significance level (in this case the significance level is equal to 0.01). If the p-value is greater than the significance level, this means that all results of the pre-test and the post-test are all hazardous values and do not have any meaning or significance. Otherwise, if the p-value is smaller than the significance level, this means that all results of both tests have a significance meaning with no chance of hazardous values. After evaluating the obtained results with the Wilcoxon test, we have obtained a p-value that is smaller than the defined significance level ($p \leq 0.01$) for phase 1 and phase 2 of this experimental study, Therefore, the results are significant at $p \leq 0.01$.

From the table 2, we conclude that both methods of learning (traditional one and by using the computer game) have a positive impact on learners' level of knowledge. In addition, the table shows that the effectiveness of using the educational computer game is slightly better compared the traditional learning. In fact, the average column of the table 2 shows that, in average, participants of both first and second group have indistinguishable levels of knowledge before the experimentation (5.65 for the first group and 5.25 for the

second group). However, after the experiment, the average of the participants in the second group (8.37) is greater than the average of the first group (7.85).

2) Impact of the educational computer game on the learners' satisfaction

Medians and Averages of the participants' answers to the TAM questionnaire were calculated and presented in the table3. An average close to 1 expresses that, in average, the participants are very satisfied when using the educational computer games. However, an average close to 7 expresses that, in average, participants are very dissatisfied when using the educational computer games. Furthermore, a median such as 1 indicates that most participants are very satisfied. Table 3 shows the medians and the averages for the variables U (Usefulness), EOU (Ease Of Use), ATT (Attitude toward using the system) and INT (Behavioral intention to use the system).

TABLE 3: AVERAGES AND MEDIANS OF LEARNERS SATISFACTION

	U	EOU	ATT	INT
Average	1.62	1.95	1.78	2.62
Median	1	2	1	2

Table 3 illustrates that, in average, participants are very satisfied when using this educational computer games. In fact, the averages for the variables EOU, U, ATT and INT are close to 1 and far from 7. As an example, the average of the ATT variable is 1.87. Furthermore, the medians of the all variables show that most participants are very satisfied when using the computer educational games. For instance, the median of the variable U is 1.

VII. CONCLUSION

This paper presented a new educational computer game called "LAST MISSION " and it is especially designed for learning the history of the World War 1 and the World War 2. In addition, this paper described an experimental study conducted with 36 volunteer participants for evaluating the effectiveness of this educational computer game. Results obtained showed that this game has a positive impact on learners' level of knowledge and learners' satisfaction.

The results obtained are in line with several studies [18, 19, 20] proved that computer games are hugely successful in engaging and motivating learners. These studies proved that these educational computer games could be an effective way to impart knowledge and provide effective learning opportunities for learners.

Although the results of this study are promising to prove the beneficial effects of this computer educational game learners' satisfaction and learners' level of knowledge, this research has to be further validated as the present samples were limited. Thus, future work will amplify and extend this study by broaden the samples.

Furthermore, the experiment of this study only used one pre-test and post-test for each learning method. This could be considered as a restriction for the generalization of the results.

Therefore, additional experiments sessions are needed with more learning contents, with longer learning sessions and tests to discover the full effect of this educational computer game.

Future works will study the potentiality of this educational computer game of extracting several learners' characteristics while playing this game. In fact, the use computer games for this purpose provides enough opportunities for learner's interactivity with the computer, which can be used to build reliable learner models.

APPENDIX

Table 4 shows the TAM questionnaire developed for determining the learners' satisfaction. In addition the table shows the distribution of learners answers.

TABLE 4: TAM QUESTIONNAIRE

Factor	Item	Answers						
		1	2	3	4	5	6	7
Usefulness (U)	1. Using LAST MISSION enhances the learning process.	11	5	0	0	0	0	0
	2. Using LAST MISSION improves the learning effectiveness.	6	7	1	2	0	0	0
	3. Overall, I find LAST MISSION useful for learning a new course.	8	7	0	1	0	0	0
Ease Of use (EOU)	4. Learning using LAST MISSION is easy for me.	6	8	1	1	0	0	0
	5. I find it takes a lot of effort to learn using LAST MISSION.	7	1	5	1	0	2	0
	6. Overall, I find LAST MISSION easy to use.	9	5	2	0	0	0	0
Attitude toward using the system (ATT)	7. I dislike the idea of using LAST MISSION.	7	7	0	2	0	0	0
	8. I have a generally favorable attitude toward using LAST MISSION.	5	7	3	1	0	0	0
	9. I believe it would be a good idea to use LAST MISSION for learning.	8	3	3	2	0	0	0
	10. Using LAST MISSION is a foolish idea.	13	1	1	1	0	0	0
Behavioral intention to use (INT)	11. I intend to use LAST MISSION during my learning process.	6	3	4	3	0	0	0
	12. I will return to LAST MISSION often.	0	6	5	3	1	1	0
	13. I intent to use LAST MISSION frequently for learning.	5	4	3	3	0	1	0

REFERENCES

- [1] Y. Bouzid, MA, Khenissi, F. Essalmi and M. Jemni, "Using educational games for sign language learning-a signwriting learning game: Case study", *Journal of Educational Technology & Society*. 19(1):129-41, 2016;
- [2] M. A. Khenissi, F. Essalmi, and M. Jemni, "A Learning version of Pacman Game," *Proc. of the Fourth International Conference On Information and Communication Technology and Accessibility*, 2013, Hammamet, Tunisia, 24-26 October 2013
- [3] M. A. Khenissi, F. Essalmi, and M. Jemni, "Comparison between serious games and learning version of existing games," *Proc. of the 6th World Conference on Educational Sciences*, 2014, 06-09, University of Malta, Malta, February 2014,
- [4] M. A. Khenissi, F. Essalmi, M. Jemni, and Kinshuk, "Review of Learner Modeling Using Educational Games," *Proc. of the 6th International Conference on Computer Supported Education*, 2014, 1-3 April 2014, Spain.
- [5] M. Papastergiou, "Digital game-based learning in high school computer science education: Impact on educational effectiveness and student motivation", *Computers & Education*, 52(1), 1-12, 2009.
- [6] J. P. Gee, "Are video games good for learning? *Nordic Journal of Digital Literacy*", 1(3), 172-182, 2006
- [7] C. Roger, "Les jeux et les hommes", Gallimard, 1957
- [8] J. H. G. Klabbers, "Three easy pieces: a taxonomy on gaming", In Saunders, D. & Severn, J. (Eds) *The International Simulation & Gaming Yearbook, Volume 7: Simulations and Games for Strategy and Policy Planning*. London: Kogan Page, 1999.
- [9] K. Oxland, "Gameplay and Design", Harlow: Addison-Wesley, 2004
- [10] R. Koster, "A Theory of Fun for Game Design", Scottsdale, Arizona: Paraglyph Press, 2005.
- [11] E. Jacques, L. Cathy, and L. Phil, "Homo ludens revisited." *Yale French Studies* 31-57, 1968.
- [12] R. Van Eck, "Digital game-based learning: It's not just the digital natives who are restless", *EDUCAUSE review*, 16, 41(2):16, 2006.
- [13] K. Kiili and K. Harri, "Exploring the learning mechanism in educational games", *In-formation Technology Interfaces*, ITI 2007. 29th International Conference on. IEEE, 2007.
- [14] C. Bisson and J. Luckner, "Fun in Learning: The Pedagogical Role of Fun in Adventure Education. Perspectives," *Journal of Experiential Education*, vol. 19, no. 2, pp. 108-112, 1996.
- [15] J. P. Gee, "High Score Education: Games, not school, are teaching kids to think", *Wired*, 11 (5), 2003
- [16] J. Kirriemuir & A. Mcfarlane, "Literature Review in Games and Learning", 2004.
- [17] A. Amory, K. Naicker, J. Vlincent, and C. Adams, C, "The use of computer games as an educational tool: Identification of appropriate game types and elements", *British J. of Educational Technology*. 30, 4(c), 311-321, 1999.
- [18] M. A. Khenissi, F. Essalmi, and M. Jemni, "A Learning version of memory match game", In 2014 IEEE 14th International Conference on Advanced Learning Technologies (pp. 209-210). IEEE., 2014.
- [19] M. A. Khenissi, F. Essalmi, and M. Jemni, "Presentation of a learning game for programming languages education", In 2013 IEEE 13th International Conference on Advanced Learning Technologies (pp. 324-326). IEEE, 2013
- [20] M. A. Khenissi, Y. Bouzid, F. Essalmi, and M. Jemni, "A learning game for deaf learners", In 2015 IEEE 15th International Conference on Advanced Learning Technologies (pp. 418-422). IEEE. 2015.
- [21] M. A. Khenissi, F. Essalmi, and M. Jemni, "Toward the personalization of learning games according to learning styles", In 2013 International Conference on Electrical Engineering and Software Applications (pp. 1-6). IEEE, 2013
- [22] P. Deubel, "Game on!", *T.H.E. Journal (Technological Horizons in Education)* 33 (6), pp 32, 2006.
- [23] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology", *MIS Quarterly*, pp. 319- 340, 1989.
- [24] E. A. Gehan, "A generalized Wilcoxon test for comparing arbitrarily singly-censored samples", *Biometrika*, 52(1-2), 203-224, 1965.

[1] Y. Bouzid, MA, Khenissi, F. Essalmi and M. Jemni, "Using educational games for sign language learning-a signwriting learning